

Cross Currency Basis Swap Pricing

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MHI

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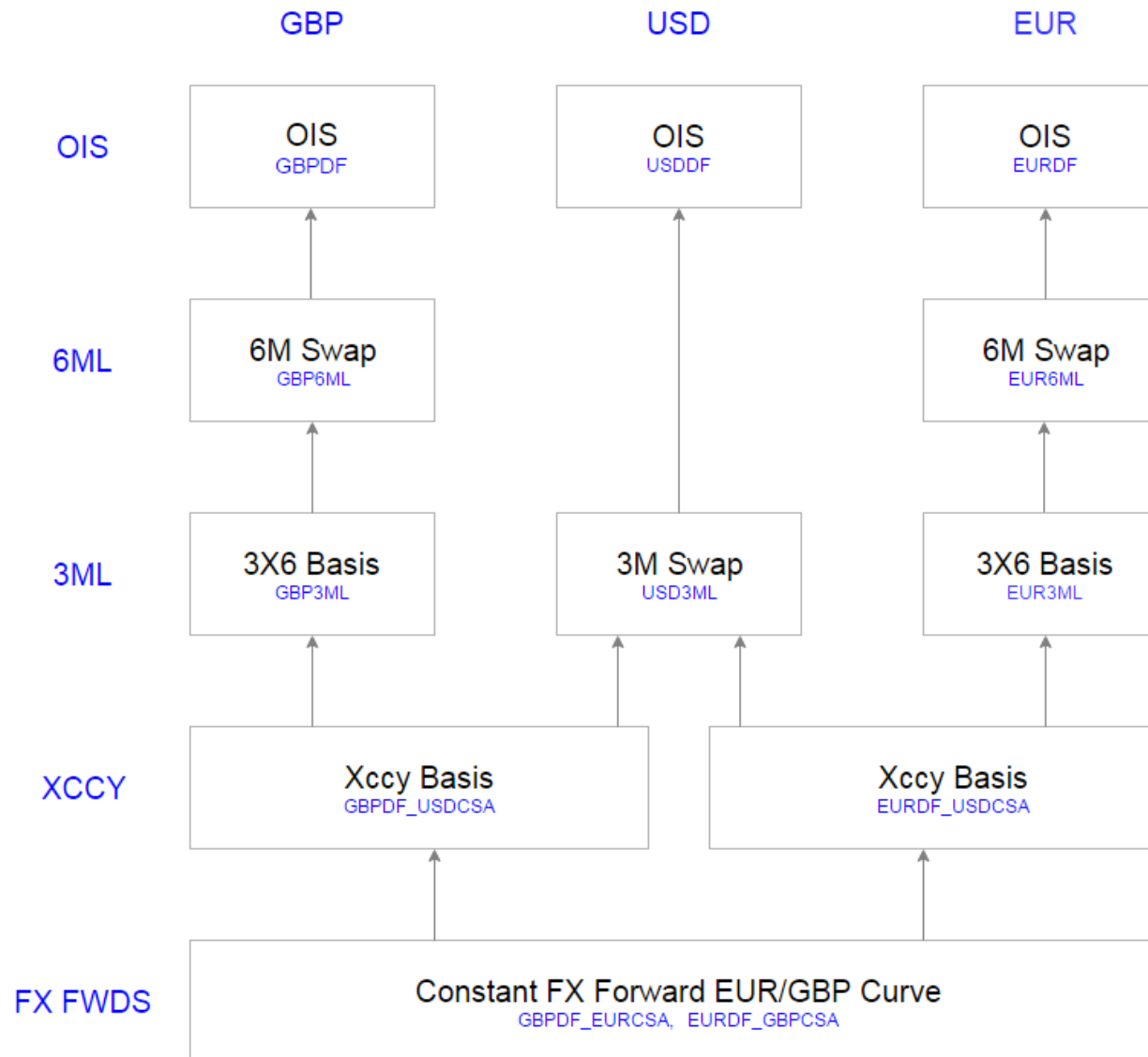
Xccy Basis Swaps

- Exchange a set of cashflows for an equivalent set in another currency
- Used to secure cheaper funding in a different currency, manage FX exposures and liquidity risk

Features

- Marked-to-Market (MTM)
- Spot & Forward FX Required
- Choice of Valuation Currency often USD
- Notional Resets on Valuation Leg
- Mitigates Basis Risk & Reduces Credit Exposure
- Multi-Currency Yield Curves Required
- OIS Discounting with Foreign CSA Collateral
- Xccy Par-Spread usually on Non-USD leg

- Consider a GBP/EUR Xccy Swap many Yield Curves are Required



Technical Notes

- MLIB supports Multi-Curve Construction with CSA Support
- Yield Curves were discussed at length in our previous Workshop
- See X:/Secure/DIR_MLib_Analytics/Workshop/Workshops_2016-10
- For CSA Calculations we assume FX Forward Invariance

FX Forward Invariance Example for EUR_JPYCSA

$$FwdFX(t, T)^{EUR/JPY} = S \underbrace{\frac{P(t, T)^{EUR_USDCSA}}{P(t, T)^{JPY_USDCSA}}}_{\text{From Xccy Curve}} = S \underbrace{\frac{P(t, T)^{EUR_JPYCSA}}{P(t, T)^{JPY_JPYCSA}}}_{\text{JPY_JPYCSA from OIS Curve}}$$

where t denotes the valuation date, S the FX spot rate, $P(t, T)$ the discount factor at time t for tenor T with $0 \leq t \leq T$

- Libor Forecast Rates do not depend on CSA Collateral

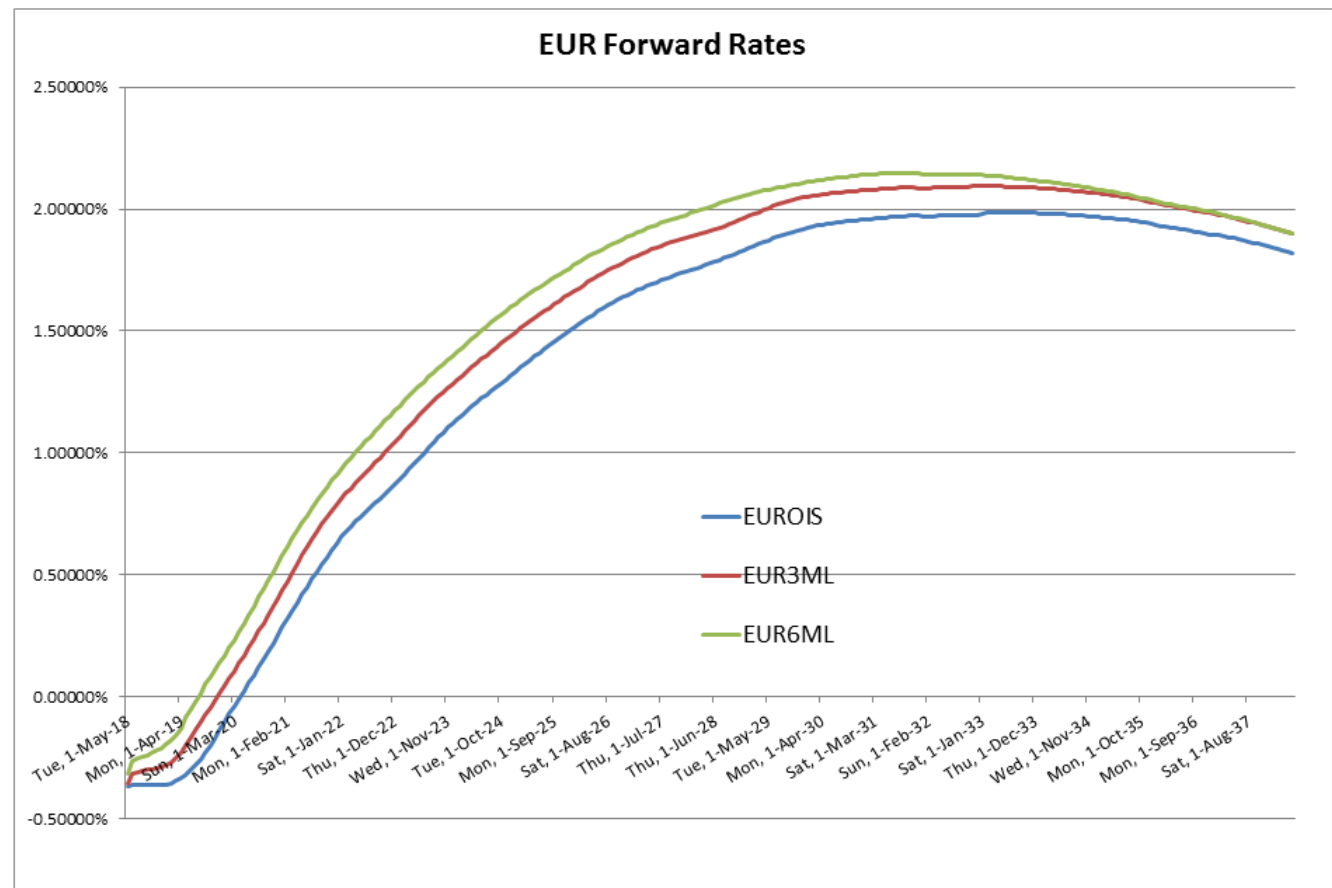
meCurveForwardRates

FixingDate	EUROIS	EUR3ML	EUR6ML
Tue, 22-May-18	-0.36600%	-0.35627%	-0.31458%
Fri, 22-Jun-18	-0.36100%	-0.31470%	-0.26573%
Mon, 23-Jul-18	-0.36100%	-0.30958%	-0.25424%
Thu, 23-Aug-18	-0.36000%	-0.30446%	-0.24626%
Mon, 24-Sep-18	-0.36000%	-0.29970%	-0.23800%

meLWOCurveForwardRatesTable

Start	0Y
End	20Y

Fixing Dates	EUROIS	EUR3ML	EUR6ML
Tue, 22-May-18	-0.36600%	-0.35627%	-0.31458%
Fri, 22-Jun-18	-0.36100%	-0.31470%	-0.26573%
Mon, 23-Jul-18	-0.36100%	-0.30958%	-0.25424%
Wed, 22-Aug-18	-0.36000%	-0.30463%	-0.24652%
Mon, 24-Sep-18	-0.36000%	-0.29970%	-0.23800%
Mon, 22-Oct-18	-0.36000%	-0.29672%	-0.22867%
Thu, 22-Nov-18	-0.35900%	-0.29342%	-0.21921%
Mon, 24-Dec-18	-0.35900%	-0.28872%	-0.20900%
Tue, 22-Jan-19	-0.35900%	-0.27789%	-0.19497%
Fri, 22-Feb-19	-0.35425%	-0.26632%	-0.17769%
Fri, 22-Mar-19	-0.34670%	-0.25430%	-0.15725%
Mon, 22-Apr-19	-0.33504%	-0.22671%	-0.13167%
Wed, 22-May-19	-0.31968%	-0.20000%	-0.08717%
Mon, 24-Jun-19	-0.29921%	-0.16689%	-0.04993%
Mon, 22-Jul-19	-0.27952%	-0.13725%	-0.01907%
Thu, 22-Aug-19	-0.25524%	-0.10516%	0.01511%
Mon, 23-Sep-19	-0.22764%	-0.07332%	0.05039%
Tue, 22-Oct-19	-0.20063%	-0.04523%	0.08236%
Fri, 22-Nov-19	-0.16863%	-0.01532%	0.11119%



- Discount Factors depend on CSA Collateral

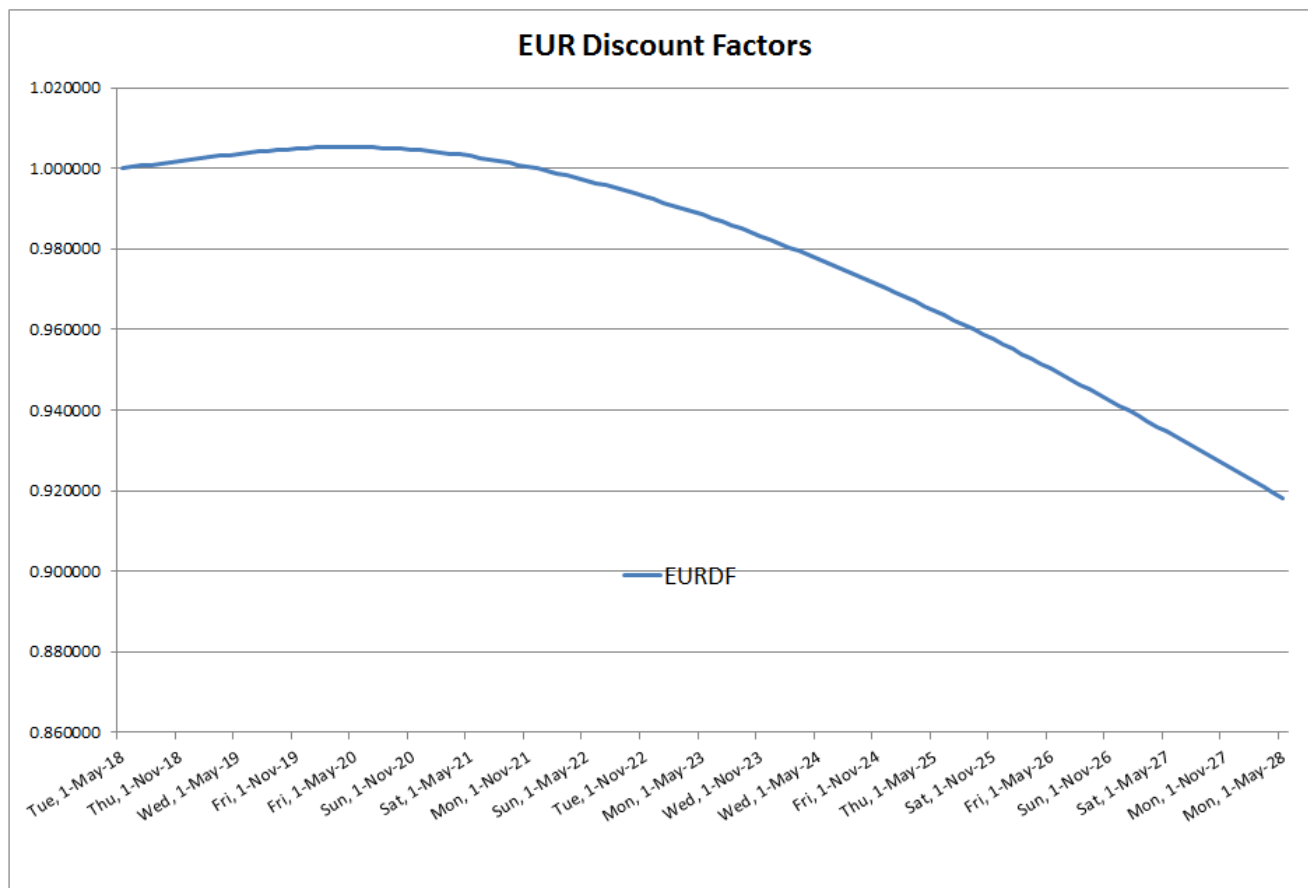
meCurveDiscountFactors

PaymentDate	EURDF
Tue, 22-May-18	1.000000
Fri, 22-Jun-18	1.000311
Mon, 23-Jul-18	1.000622
Thu, 23-Aug-18	1.000933
Mon, 24-Sep-18	1.001253

meLWOCurveDiscountFactorsTable

Start	OD
End	10Y

Payment Dates	EURDF
Tue, 22-May-18	1.000000
Fri, 22-Jun-18	1.000311
Sun, 22-Jul-18	1.000612
Wed, 22-Aug-18	1.000923
Sat, 22-Sep-18	1.001233
Mon, 22-Oct-18	1.001533
Thu, 22-Nov-18	1.001843
Sat, 22-Dec-18	1.002143
Tue, 22-Jan-19	1.002453
Fri, 22-Feb-19	1.002762
Fri, 22-Mar-19	1.003035
Mon, 22-Apr-19	1.003330
Wed, 22-May-19	1.003605
Sat, 22-Jun-19	1.003873
Mon, 22-Jul-19	1.004116
Thu, 22-Aug-19	1.004348
Sun, 22-Sep-19	1.004558
Tue, 22-Oct-19	1.004738
Fri, 22-Nov-19	1.004899
Sun, 22-Dec-19	1.005028



USD 10MM 5Y EUR/USD Float-Float Xccy Swap

- USD 10MM
- EUR 8.6MM
- USD3ML vs EUR3ML-15.3 bps
- EUR Float Spread
- Marked-to-Market
- USD Notional Reset
- USD Valuation
- USD Collateral

30) Solver (Premium)				31) Load		32) Save		35) Trade		38) CCP							
3) Main		4) Details		5) Curves		6) Cashflow		7) Resets		9) Scenario		10) Risk		11) CVA		12) Matrix	
Deal		MTM XCCY Swap		Counterparty		SWAP CNTRPARTY		+ Ticker / SWAP		20) Properties							
Swap		*Notional Reset b...				3 Month Euribor		Valuation Settings									
Leg 1:Float		Receive		Leg 2:Float		Pay		Curve Date		07/26/2018							
Notional		10MM		Notional		8,562,596.86		Valuation		07/30/2018							
Currency		USD		Currency		EUR		CSA Coll Ccy		USD							
Effective		0D 07/30/2018		Effective		0D 07/30/2018		Valuation Ccy		USD							
Maturity		5Y 07/30/2023		Maturity		5Y 07/30/2023		FX Rate		1.167870							
Index		3M US0003M		Index		3M EUR003M		☒ OIS DC Stripping									
Spread		0.000 bp		Spread		-15.312 bp											
Leverage		1.00000		Leverage		1.00000											
Latest Index		2.33888		Latest Index		-0.32000											
Reset Freq		Quarterly		Reset Freq		Quarterly											
Pay Freq		Quarterly		Pay Freq		Quarterly											
Day Count		ACT/360		Day Count		ACT/360											
Market																	
Leg 1: NPV		10,156,257.24		Leg 2: NPV		-10,156,257.24											
Accrued		0.00		Accrued		0.00											
Premium		101.56		Premium		-101.56											
DV01		33.01		DV01		-33.01											
Valuation Results												22) Calculators*					
Principal		0.00		Premium		0.00000		BR01 92:EUR vs.		-5,134.54							
Accrued		0.00		BP Value		0.00000		DV01		0.00							
NPV		0.00						Gamma (1bp)		0.00							

USD Discounting: USDOIS

EUR Discounting: EUROIS_USDCSA

USD Cashflows

- USD 10MM with Notional Reset
- Floating Index: USD3ML Flat
- Discounting using USDOIS Curve

30) Solver (Premium)				31) Load		32) Save		35) Trade		38) CCP	
3) Main	4) Details	5) Curves	6) Cashflow	7) Resets	9) Scenario	10) Risk	11) CVA	12) Matrix			
21) Cashflow Table		22) Cashflow Graph									
Cashflow	Leg 1: Receive Float			Historical Cashflows				Accrued		0.00	
Currency	USD			Zero Rate				NPV		10,156,257.24	
				Equiv. Coupon							
 											
Pay Date	Accrual Start	Accrual End	Days	FX Fixing Date	FX Rate	Notional	Principal	Reset Date			
04/30/2019	01/30/2019	04/30/2019	90	01/28/2019	0.84400	10,145,201.92	-77,656.99	01/28/2019			
07/30/2019	04/30/2019	07/30/2019	91	04/26/2019	0.83759	10,222,858.91	-84,035.11	04/26/2019			
10/30/2019	07/30/2019	10/30/2019	92	07/26/2019	0.83076	10,306,894.02	-88,343.61	07/26/2019			
01/30/2020	10/30/2019	01/30/2020	92	10/28/2019	0.82370	10,395,237.63	-86,616.99	10/28/2019			
04/30/2020	01/30/2020	04/30/2020	91	01/28/2020	0.81690	10,481,854.62	-85,829.80	01/28/2020			
07/30/2020	04/30/2020	07/30/2020	91	04/28/2020	0.81026	10,567,684.43	-85,045.73	04/28/2020			
10/30/2020	07/30/2020	10/30/2020	92	07/28/2020	0.80379	10,652,730.16	-84,097.09	07/28/2020			
01/29/2021	10/30/2020	01/29/2021	91	10/28/2020	0.79750	10,736,827.25	-81,143.43	10/28/2020			
04/30/2021	01/29/2021	04/30/2021	91	01/27/2021	0.79152	10,817,970.68	-79,217.28	01/27/2021			
07/30/2021	04/30/2021	07/30/2021	91	04/28/2021	0.78576	10,897,187.95	-77,398.69	04/28/2021			
10/29/2021	07/30/2021	10/29/2021	91	07/28/2021	0.78022	10,974,586.64	-75,689.19	07/28/2021			
01/31/2022	10/29/2021	01/31/2022	94	10/27/2021	0.77488	11,050,275.83	-74,854.82	10/27/2021			
04/29/2022	01/31/2022	04/29/2022	88	01/27/2022	0.76966	11,125,130.65	-71,659.94	01/27/2022			
07/29/2022	04/29/2022	07/29/2022	91	04/27/2022	0.76474	11,196,790.59	-70,936.60	04/27/2022			
10/31/2022	07/29/2022	10/31/2022	94	07/27/2022	0.75992	11,267,727.19	-70,215.14	07/27/2022			
01/30/2023	10/31/2022	01/30/2023	91	10/27/2022	0.75522	11,337,942.33	-68,001.19	10/27/2022			
04/28/2023	01/30/2023	04/28/2023	88	01/26/2023	0.75071	11,405,943.51	-65,869.32	01/26/2023			
07/31/2023	04/28/2023	07/31/2023	94	04/26/2023	0.74640	11,471,812.84	11,471,812.84	04/26/2023			

EUR Cashflows

- EUR 8.6MM
- Floating Index EUR3ML - 15.3 bps (Par Spread)
- FX Adjustment for Valuation in USD
- Discounting using EUR_USDCSA Curve

30) Solver (Premium)				31) Load		32) Save		35) Trade		38) CCP	
3) Main	4) Details	5) Curves	6) Cashflow	7) Resets	9) Scenario	10) Risk	11) CVA	12) Matrix			
21) Cashflow Table		22) Cashflow Graph									
Cashflow		Leg 2: Pay Float		■ Historical Cashflows		Accrued		0.00			
Currency		EUR		■ Zero Rate		NPV		-8,696,393.64			
				■ Equiv. Coupon							
Pay Date	Accrual Start	Accrual End	Days	Notional	Principal	Reset Date	Reset Rate	Payment			
04/30/2019	01/30/2019	04/30/2019	90	-8,562,596.86	0.00	01/28/2019	-0.29252	9,539.71			
07/30/2019	04/30/2019	07/30/2019	91	-8,562,596.86	0.00	04/26/2019	-0.27338	9,231.48			
10/30/2019	07/30/2019	10/30/2019	92	-8,562,596.86	0.00	07/26/2019	-0.24222	8,650.94			
01/30/2020	10/30/2019	01/30/2020	92	-8,562,596.86	0.00	10/28/2019	-0.16018	6,855.88			
04/30/2020	01/30/2020	04/30/2020	91	-8,562,596.86	0.00	01/28/2020	-0.07962	5,037.59			
07/30/2020	04/30/2020	07/30/2020	91	-8,562,596.86	0.00	04/28/2020	0.00486	3,209.14			
10/30/2020	07/30/2020	10/30/2020	92	-8,562,596.86	0.00	07/28/2020	0.08914	1,400.11			
01/29/2021	10/30/2020	01/29/2021	91	-8,562,596.86	0.00	10/28/2020	0.17463	-465.55			
04/30/2021	01/29/2021	04/30/2021	91	-8,562,596.86	0.00	01/27/2021	0.25996	-2,312.40			
07/30/2021	04/30/2021	07/30/2021	91	-8,562,596.86	0.00	04/28/2021	0.34554	-4,164.64			
10/29/2021	07/30/2021	10/29/2021	91	-8,562,596.86	0.00	07/28/2021	0.43036	-6,000.65			
01/31/2022	10/29/2021	01/31/2022	94	-8,562,596.86	0.00	10/27/2021	0.51171	-8,017.24			
04/29/2022	01/31/2022	04/29/2022	88	-8,562,596.86	0.00	01/27/2022	0.58596	-9,059.53			
07/29/2022	04/29/2022	07/29/2022	91	-8,562,596.86	0.00	04/27/2022	0.65336	-10,827.32			
10/31/2022	07/29/2022	10/31/2022	94	-8,562,596.86	0.00	07/27/2022	0.71814	-12,632.52			
01/30/2023	10/31/2022	01/30/2023	91	-8,562,596.86	0.00	10/27/2022	0.78157	-13,602.33			
04/28/2023	01/30/2023	04/28/2023	88	-8,562,596.86	0.00	01/26/2023	0.84255	-14,430.22			
07/31/2023	04/28/2023	07/31/2023	94	-8,562,596.86	-8,562,596.86	04/26/2023	0.90425	-8,579,390.40			

USD 10MM 5Y EUR/USD Float-Float Xccy Swap

- USD 10MM
- EUR 8.6MM
- USD3ML vs EUR3ML-15.5 bps
- EUR Float Spread
- Marked-to-Market
- USD Notional Reset
- USD Valuation
- USD Collateral

Create Xccy Swap(s)		XCCY1@283:6
Leverage	1.00	1.00
Currency	EUR	USD
Notional / Ccy	10,000,000	USD
Effective	2D	Mon, 30-Jul-18
Maturity	5Y	Sun, 30-Jul-23
Valuation Ccy	USD	
FX (real time)	1.1660	
FX (override)		
Notional Reset	Y	USD
Pay/Receive	PAY	RECEIVE
Leg type	FLOATING	FLOATING
Index	EUR EURIBOR 3M	USD LIBOR 3M
Rate(%) / Margin(bps)	-15.500	0.0
Frequency	QUARTERLY	QUARTERLY
CSA	USDCSA	
Solver Leg	Left	

Fees		Currency	USD
		Pay/Receive	RECEIVE
		Amount	0
		Payment Date	Mon, 30-Jul-18

Prices		USD	
SwapPV		0	
ParSpread	0.000		
LegPV	-141,459	141,459	
	EUR	USD	

EURUSD Xccy basis	
3M	-0.040000%
6M	-0.113750%
9M	-0.110000%
1Y	-0.110000%
2Y	-0.115000%
3Y	-0.125000%
4Y	-0.140000%
5Y	-0.155000%
6Y	-0.165000%
7Y	-0.172500%
8Y	-0.176250%
9Y	-0.177500%
10Y	-0.176250%
12Y	-0.168750%
15Y	-0.142500%
20Y	-0.091250%
25Y	-0.040000%
30Y	-0.007500%

USD Discounting: USDOIS

EUR Discounting: EUROIS_USDCSA

USD 10MM 5Y EUR/USD Float-Float Xccy Swap

SwapLegsLVB

LegType	Leg1:FLOAT	Leg2:FLOAT
Notional	8,582,954	10,000,000
NotionalExchange	START_AND_END	START_AND_END
PayReceive	PAYER	RECEIVER
ForecastCurve	EUR3ML	USD3ML
DiscountCurve	EURDF_USDCSA	USDDF_NOCSA
Currency	EUR	USD
Leverage	1.00	1.00
FixedRate		
FirstStubCurveIndex	NATURAL	NATURAL
LastStubCurveIndex	NATURAL	NATURAL
FirstFixing		
LastFixing		
FloatSpread	-15.5	0.0
EffectiveDate	Tue, 31-Jul-18	Tue, 31-Jul-18
MaturityDate	Mon, 31-Jul-23	Mon, 31-Jul-23
Frequency	QUARTERLY	QUARTERLY
Daycount	ACT/360	ACT/360
FixingBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING
FixingCalendar	TGT:NYB:LNb	TGT:NYB:LNb
AccrualBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING
AccrualCalendar	TGT:NYB:LNb	TGT:NYB:LNb
PaymentBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING
PaymentCalendar	TGT:NYB:LNb	TGT:NYB:LNb
FirstStubDate		
LastStubDate		
RollDay	NORMAL	NORMAL
FixingLag	0D	0D
FirstFixingLag	0D	0D
PaymentLag	0D	0D
StubType	ShortStart	ShortStart

SwapPropertiesLVB

MTM	TRUE
ValuationCurrency	USD
FxRate	1.16510
NotionalFxResetLeg	Leg2:FLOAT

Must Quote FX Spot using Valuation Currency as Foreign Currency, in this case as EUR/USD

USD Cashflows

- USD 10MM with Notional Reset and Notional Exchange
- Floating Index: USD3ML Flat
- Discounting using USDOIS Curve

FixingDate	AccrualStart	AccrualEnd	AccrualDays	AccrualYearFractions	PaymentDate	Notional	NotionalExchange	Leverage	FloatSpread	FloatRate	FxFixingDate	FxRate	rvivalProbabil	Coupon	RiskyCoupon	DiscountFactor	CouponPV
					Mon, 30-Jul-18		-10,002,877	1.00					1	-10,002,877	-10,002,877	0.999787	-10,000,747
Mon, 30-Jul-18	Mon, 30-Jul-18	Tue, 30-Oct-18	92	0.26	Tue, 30-Oct-18	10,002,877	-69,055	1.00	0.00	2.33888%	Mon, 30-Jul-18	0.85746	1	-9,266	-9,266	0.994640	-9,217
Tue, 30-Oct-18	Tue, 30-Oct-18	Wed, 30-Jan-19	92	0.26	Wed, 30-Jan-19	10,071,932	-78,577	1.00	0.00	2.53751%	Tue, 30-Oct-18	0.85158	1	-13,263	-13,263	0.988925	-13,117
Wed, 30-Jan-19	Wed, 30-Jan-19	Tue, 30-Apr-19	90	0.25	Tue, 30-Apr-19	10,150,509	-79,748	1.00	0.00	2.73330%	Wed, 30-Jan-19	0.84499	1	-10,387	-10,387	0.982855	-10,209
Tue, 30-Apr-19	Tue, 30-Apr-19	Tue, 30-Jul-19	91	0.25	Tue, 30-Jul-19	10,230,257	-83,632	1.00	0.00	2.86822%	Tue, 30-Apr-19	0.83840	1	-9,460	-9,460	0.976698	-9,240
Tue, 30-Jul-19	Tue, 30-Jul-19	Wed, 30-Oct-19	92	0.26	Wed, 30-Oct-19	10,313,889	-85,160	1.00	0.00	2.95804%	Tue, 30-Jul-19	0.83160	1	-7,193	-7,193	0.970270	-6,979
Wed, 30-Oct-19	Wed, 30-Oct-19	Thu, 30-Jan-20	92	0.26	Thu, 30-Jan-20	10,399,049	-85,557	1.00	0.00	3.01098%	Wed, 30-Oct-19	0.82479	1	-5,539	-5,539	0.963575	-5,337
Thu, 30-Jan-20	Thu, 30-Jan-20	Thu, 30-Apr-20	91	0.25	Thu, 30-Apr-20	10,484,606	-84,873	1.00	0.00	3.03271%	Thu, 30-Jan-20	0.81806	1	-4,498	-4,498	0.956791	-4,303
Thu, 30-Apr-20	Thu, 30-Apr-20	Thu, 30-Jul-20	91	0.25	Thu, 30-Jul-20	10,569,479	-84,565	1.00	0.00	3.03220%	Thu, 30-Apr-20	0.81149	1	-3,553	-3,553	0.949983	-3,375
Thu, 30-Jul-20	Thu, 30-Jul-20	Fri, 30-Oct-20	92	0.26	Fri, 30-Oct-20	10,654,044	-84,017	1.00	0.00	3.01431%	Thu, 30-Jul-20	0.80505	1	-1,946	-1,946	0.943242	-1,836
Fri, 30-Oct-20	Fri, 30-Oct-20	Fri, 29-Jan-21	91	0.25	Fri, 29-Jan-21	10,738,061	-81,050	1.00	0.00	2.97417%	Fri, 30-Oct-20	0.79875	1	-321	-321	0.936750	-301
Fri, 29-Jan-21	Fri, 29-Jan-21	Fri, 30-Apr-21	91	0.25	Fri, 30-Apr-21	10,819,111	-78,909	1.00	0.00	2.95722%	Fri, 29-Jan-21	0.79277	1	1,966	1,966	0.930408	1,829
Fri, 30-Apr-21	Fri, 30-Apr-21	Fri, 30-Jul-21	91	0.25	Fri, 30-Jul-21	10,898,019	-76,867	1.00	0.00	2.95476%	Fri, 30-Apr-21	0.78703	1	4,531	4,531	0.924171	4,187
Fri, 30-Jul-21	Fri, 30-Jul-21	Fri, 29-Oct-21	91	0.25	Fri, 29-Oct-21	10,974,886	-75,301	1.00	0.00	2.96749%	Fri, 30-Jul-21	0.78152	1	7,023	7,023	0.917971	6,447
Fri, 29-Oct-21	Fri, 29-Oct-21	Mon, 31-Jan-22	94	0.26	Mon, 31-Jan-22	11,050,187	-76,187	1.00	0.00	2.97709%	Fri, 29-Oct-21	0.77619	1	9,712	9,712	0.911590	8,853
Mon, 31-Jan-22	Mon, 31-Jan-22	Fri, 29-Apr-22	88	0.24	Fri, 29-Apr-22	11,126,375	-69,794	1.00	0.00	2.97720%	Mon, 31-Jan-22	0.77088	1	11,180	11,180	0.905633	10,125
Fri, 29-Apr-22	Fri, 29-Apr-22	Fri, 29-Jul-22	91	0.25	Fri, 29-Jul-22	11,196,168	-70,572	1.00	0.00	2.97029%	Fri, 29-Apr-22	0.76607	1	13,492	13,492	0.899487	12,136
Fri, 29-Jul-22	Fri, 29-Jul-22	Mon, 31-Oct-22	94	0.26	Mon, 31-Oct-22	11,266,740	-71,215	1.00	0.00	2.95711%	Fri, 29-Jul-22	0.76127	1	15,779	15,779	0.893158	14,093
Mon, 31-Oct-22	Mon, 31-Oct-22	Mon, 30-Jan-23	91	0.25	Mon, 30-Jan-23	11,337,955	-67,557	1.00	0.00	2.94617%	Mon, 31-Oct-22	0.75649	1	16,880	16,880	0.887095	14,974
Mon, 30-Jan-23	Mon, 30-Jan-23	Fri, 28-Apr-23	88	0.24	Fri, 28-Apr-23	11,405,512	-64,208	1.00	0.00	2.94030%	Mon, 30-Jan-23	0.75201	1	17,768	17,768	0.881349	15,660
Fri, 28-Apr-23	Fri, 28-Apr-23	Mon, 31-Jul-23	94	0.26	Mon, 31-Jul-23	11,469,720	11,469,720	1.00	0.00	2.94006%	Fri, 28-Apr-23	0.74780	1	11,557,771	11,557,771	0.875412	10,117,815

$$\text{Coupon(USD)} = \text{Notional(USD)} \times \text{USD3ML} \times \text{YearFraction}$$

$$\text{Notional(USD, } t) = \text{Notional(EUR, } t) \times \text{FXForward(EUR/USD, } t)$$

$$\text{NotionalExchange(USD, } t) = \text{Notional(USD, } t) - \text{Notional(USD, } t + 1)$$

EUR Cashflows

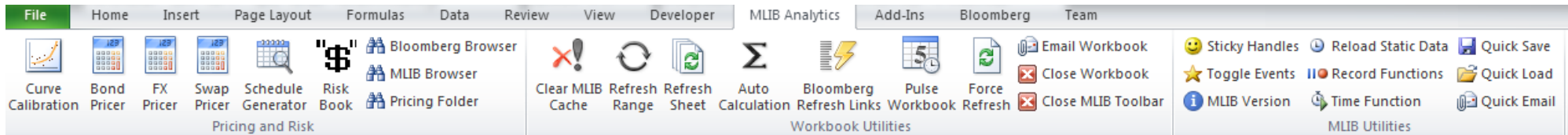
- EUR 8.6MM (Constant)
- Floating Index EUR3ML - 15.5 bps (Par Spread)
- FX Adjustment for Valuation in USD
- Discounting using EUR_USDCSA Curve

LEG1:FLOAT Leg Sci															
FixingDate	AccrualStart	AccrualEnd	AccrualDays	AccrualYearFractions	PaymentDate	Notional	NotionalExchange	Leverage	FloatSpread	FloatRate	SurvivalProbability	DefaultProb	Coupon	DiscountFactor	CouponPV
Mon, 30-Jul-18	Mon, 30-Jul-18	Tue, 30-Oct-18	92	0.26	Mon, 30-Jul-18	-8,577,065	8,577,065	1.000	-15.50	-0.32000%	1.00	0.00	8,577,065	1.000075	10,000,747
Tue, 30-Oct-18	Tue, 30-Oct-18	Wed, 30-Jan-19	92	0.26	Tue, 30-Oct-18	-8,577,065	0	1.000	-15.50	-0.32396%	1.00	0.00	10,412	1.001795	12,161
Wed, 30-Jan-19	Wed, 30-Jan-19	Tue, 30-Apr-19	90	0.25	Wed, 30-Jan-19	-8,577,065	0	1.000	-15.50	-0.30379%	1.00	0.00	9,838	1.005486	11,533
Tue, 30-Apr-19	Tue, 30-Apr-19	Tue, 30-Jul-19	91	0.25	Tue, 30-Apr-19	-8,577,065	0	1.000	-15.50	-0.25285%	1.00	0.00	8,843	1.007355	10,385
Tue, 30-Jul-19	Tue, 30-Jul-19	Wed, 30-Oct-19	92	0.26	Tue, 30-Jul-19	-8,577,065	0	1.000	-15.50	-0.17688%	1.00	0.00	7,274	1.008989	8,558
Wed, 30-Oct-19	Wed, 30-Oct-19	Thu, 30-Jan-20	92	0.26	Wed, 30-Oct-19	-8,577,065	0	1.000	-15.50	-0.10244%	1.00	0.00	5,643	1.010270	6,647
Thu, 30-Jan-20	Thu, 30-Jan-20	Thu, 30-Apr-20	91	0.25	Thu, 30-Jan-20	-8,577,065	0	1.000	-15.50	-0.03717%	1.00	0.00	4,166	1.011278	4,912
Thu, 30-Apr-20	Thu, 30-Apr-20	Thu, 30-Jul-20	91	0.25	Thu, 30-Apr-20	-8,577,065	0	1.000	-15.50	0.01904%	1.00	0.00	2,948	1.012116	3,479
Thu, 30-Jul-20	Thu, 30-Jul-20	Fri, 30-Oct-20	92	0.26	Thu, 30-Jul-20	-8,577,065	0	1.000	-15.50	0.07115%	1.00	0.00	1,838	1.012859	2,170
Fri, 30-Oct-20	Fri, 30-Oct-20	Fri, 29-Jan-21	91	0.25	Fri, 30-Oct-20	-8,577,065	0	1.000	-15.50	0.13521%	1.00	0.00	429	1.013481	507
Fri, 29-Jan-21	Fri, 29-Jan-21	Fri, 30-Apr-21	91	0.25	Fri, 29-Jan-21	-8,577,065	0	1.000	-15.50	0.21464%	1.00	0.00	-1,293	1.013960	-1,529
Fri, 30-Apr-21	Fri, 30-Apr-21	Fri, 30-Jul-21	91	0.25	Fri, 30-Apr-21	-8,577,065	0	1.000	-15.50	0.31069%	1.00	0.00	-3,376	1.014267	-3,992
Fri, 30-Jul-21	Fri, 30-Jul-21	Fri, 29-Oct-21	91	0.25	Fri, 30-Jul-21	-8,577,065	0	1.000	-15.50	0.41724%	1.00	0.00	-5,686	1.014375	-6,724
Fri, 29-Oct-21	Fri, 29-Oct-21	Mon, 31-Jan-22	94	0.26	Fri, 29-Oct-21	-8,577,065	0	1.000	-15.50	0.51692%	1.00	0.00	-8,105	1.014269	-9,585
Mon, 31-Jan-22	Mon, 31-Jan-22	Fri, 29-Apr-22	88	0.24	Mon, 31-Jan-22	-8,577,065	0	1.000	-15.50	0.60176%	1.00	0.00	-9,367	1.013962	-11,073
Fri, 29-Jul-22	Fri, 29-Jul-22	Mon, 31-Oct-22	94	0.26	Fri, 29-Apr-22	-8,577,065	0	1.000	-15.50	0.73418%	1.00	0.00	-12,971	1.012659	-15,315
Mon, 31-Oct-22	Mon, 31-Oct-22	Mon, 30-Jan-23	91	0.25	Mon, 31-Oct-22	-8,577,065	0	1.000	-15.50	0.79486%	1.00	0.00	-13,873	1.011777	-16,365
Mon, 30-Jan-23	Mon, 30-Jan-23	Fri, 28-Apr-23	88	0.24	Mon, 30-Jan-23	-8,577,065	0	1.000	-15.50	0.85365%	1.00	0.00	-14,648	1.010883	-17,264
Fri, 28-Apr-23	Fri, 28-Apr-23	Mon, 31-Jul-23	94	0.26	Fri, 28-Apr-23	-8,577,065	-8,577,065	1.000	-15.50	0.91357%	1.00	0.00	-8,594,054	1.009974	-10,119,747

$$Coupon(EUR) = Notional(EUR) \times (EUR3ML - 15.5bps) \times YearFraction$$

$$PVLet(EUR) = Coupon(EUR) \times FXSpot(EUR/USD) \times DF(EUR_USDCSA)$$

Pricer: MLIB Ribbon - Pricing Folder - Xccy Pricer



Refresh Major Currencies using Buttons 1-2 and Minors using 1-4

Valuation Date
Thu, 26-Jul-18

Stream Data
FALSE

Enable Bloomberg
TRUE

Broker Selection Profile
QUANT

Curve Calibration

1. Request Curve Data

2. Calibrate Curves

Curve Navigation

Calibration Instruments	Market Data	Curve Set-Up	Curve Construction	Static Data
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Major Currencies

1 2

Other Currency Pairs

3 4

	EUR	GBP	JPY	USD	AUD	CHF	CNH	DKK	NOK	SEK	CAD	HKD	NZD	ZAR
Enable Curves >>>	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Curve Build: Successful Curve Build	OK	OK	OK	OK	OK									
Market Data: # No. MktData Points Loading	OK	OK	OK	OK	OK									
OIS Curve Interpolation Use Libor-OIS Basis Starting Point: Libor-OIS Basis	SPLINE FALSE	SPLINE TRUE 1Y	SPLINE TRUE 1Y	SPLINE TRUE 1Y	SPLINE FALSE 30Y									
STD Swap Curve FRA Frequency Interpolation: Futures / FRAs Interpolation: Swaps Futures: HW1F Mean Reversion Futures: HW1F Volatility Apply Tension Tension Gap	6M LINEARSPLINE LINEARSPLINE FALSE 2	NONE LINEARSPLINE LINEARSPLINE 3.00% 0.50% FALSE 2	6M LINEARSPLINE LINEARSPLINE FALSE 2	NONE LINEARSPLINE LINEARSPLINE 3.00% 1.50% FALSE 2	3M LINEARSPLINE LINEARSPLINE FALSE 2									
Tenor Basis Curve(s) Interpolation: Basis	SPLINE	SPLINE	SPLINE	SPLINE	SPLINE									
Xccy and FX Fwd Constant Curve(s) Xccy curve uses FX market Interpolation: Xccy / FX Fwds	FALSE SPLINE	FALSE SPLINE	FALSE SPLINE	SPLINE	FALSE SPLINE									

Supports Pricing, Cashflow Display & Bucketed IR Delta Risk

Major Currencies
Error Count 0
More Deals
Cashflow Info

Create Xccy Swap(s)	XCCY1@249:6	XCCY2@276:6	XCCY3@266:6	XCCY4@278:6
Leverage	1.00	1.00	1.00	1.00
Currency	EUR	USD	EUR	USD
Notional / Ccy	1,000,000	USD	1,000,000	USD
Effective	2D	Mon, 30-Jul-18	2D	Mon, 30-Jul-18
Maturity	5Y	Sun, 30-Jul-23	5Y	Sun, 30-Jul-23
Valuation Ccy	USD	USD	USD	USD
FX (real time)	1.1651	1.1651	1.1651	1.1651
FX (override)				
Notional Reset	Y	USD	Y	USD
Pay/Receive	PAY	RECEIVE	PAY	RECEIVE
Leg type	FLOATING	FLOATING	FLOATING	FLOATING
Index	EUR EURIBOR 3M	USD LIBOR 3M	EUR EURIBOR 3M	USD LIBOR 3M
Rate(%) / Margin(bps)	-15.5	0.0	0.0	0.0
Frequency	QUARTERLY	QUARTERLY	QUARTERLY	QUARTERLY
CSA	USDCSA	USDCSA	USDCSA	USDCSA
Solver Leg	Left	Left	Left	Left

Fees	Currency	USD	Currency	USD	Currency	USD	Currency	USD
Pay/Receive	RECEIVE	RECEIVE	Pay/Receive	RECEIVE	Pay/Receive	RECEIVE	Pay/Receive	RECEIVE
Amount	0	0	Amount	0	Amount	0	Amount	0
PaymentDate	Mon, 30-Jul-18	Mon, 30-Jul-18	PaymentDate	Mon, 30-Jul-18	PaymentDate	Mon, 30-Jul-18	PaymentDate	Mon, 30-Jul-18

Prices	USD	USD	USD	USD
SwapPV	0	-7,952	-7,952	-7,952
ParSpread	0.000	-15.500	-15.500	-15.500
LegPV	-14,318	-22,270	-22,270	-22,270
	EUR	EUR	EUR	EUR
	USD	USD	USD	USD

USDJPY Xccy basis	EURUSD Xccy basis
3M -0.107500%	3M -0.040000%
6M -0.260000%	6M -0.113750%
9M -0.250000%	9M -0.110000%
1Y -0.257500%	1Y -0.110000%
18M -0.300000%	2Y -0.115000%
2Y -0.311250%	3Y -0.125000%
3Y -0.360000%	4Y -0.140000%
4Y -0.400000%	5Y -0.155000%
5Y -0.430000%	6Y -0.165000%
6Y -0.448750%	7Y -0.172500%
7Y -0.460000%	8Y -0.176250%
8Y -0.465000%	9Y -0.177500%
9Y -0.466250%	10Y -0.176250%
10Y -0.465000%	12Y -0.168750%
12Y -0.460000%	15Y -0.142500%
15Y -0.450000%	20Y -0.091250%
20Y -0.435000%	25Y -0.040000%
25Y -0.423750%	30Y -0.007500%
30Y -0.417500%	

The End